Supplements

Supplement S1: Supplement to the transition probability of colony strength

This section provides supplementary materials on the transition probability of colony strength, including the biological background, mathematical formulation, simulation experiments, and its derivation, as elaborated below.

S1.1 Biological background

A honeybee colony, as an intelligent superorganism with a degree of self-regulation and social resilience, maintains homeostasis despite internal and external disturbances (Ulgezen et al., 2021). The survival and development of bee colonies are highly dependent on individual quality, resilience ability, and environmental conditions. A honey bee colony typically consists of ten thousand to fifty or sixty thousand individuals, including one queen, hundreds of males (i.e., drones), and thousands of sterile females (known as worker bees). The queen lays eggs daily to produce offspring. Workers undergo several developmental stages, namely egg, larva, pupa, capped, uncapped, and adult. Upon reaching adulthood, individuals perform age-based tasks: younger individuals carry out in-hive duties, such as cleaning and brood rearing, while older individuals transition to foraging activities outside the hive, a form of social division of labor. Aged foragers face elevated external risks and physiological exhaustion, and their mortality is generally considered a natural outcome. Notably, this social division of labor can be adaptively adjusted in response to internal biological behaviors and external stressors through modifications in recruitment behavior (Khoury et al., 2011).

In addition to the age-structure and social division mechanisms, other factors also play important roles in colony dynamics, as evidenced by comprehensive research. (i) Seasonal factors: Related research revealed that egg-laying rate (Messan et al., 2021) and individual lifespan (Lemanski et al., 2020) correlate with the seasonal pattern and demonstrate the important role of seasonality in colony dynamics (Chen et al., 2020, 2023). (ii) Food/nutrition supply: Food collection depends on both the number of foragers and the collection efficiency of individuals. According to adaptive recruitment, the number of foragers evolves in response to food reserves in the hive and the number of foragers themselves, by accelerating, slowing, or even reversing recruitment behaviors (Khoury et al., 2013; Perry et al., 2015). The collection efficiency of individuals depends on environmental factors, such as floral abundance and weather conditions. Food reserves, on the one hand, support individual survival by providing sufficient energy and nutrients to bees of all ages, especially during the brood phase; on the other hand, foraging operations may affect mortality rates due to work fatigue (Hong et al.,

2022). (iii) Other stressors: Some lethal and sublethal stressors, such as parasite infection (Muhammad & Eberl, 2020), viral invasion (Kang et al., 2016), and pesticide exposure (Betti et al., 2014), are also important drivers of colony dynamics (Banks et al., 2020). Further understanding can be found in the review by Chen et al. (2021).

S1.2 Mathematical model

A bee colony consists of two age groups: brood and adult. As illustrated in Fig. S1-1, eggs hatch into brood (e.g., larva, pupa, and capped, and uncapped), whose survival is influenced by both the nursing efficiency provided by hive bees and food reserves within the hive. The brood population subsequently develops into adult bees. A portion of the hive bees are recruited into the forager population, while foragers may revert to the hive class through social inhibition. The recruitment process might be accelerated, slowed, or even reversed, depending on food availability and active foragers. Additionally, food collection by foragers is influenced by the number of foragers and external conditions, such as floral abundance and habitat suitability. The collected food supplies provide necessary energy to the brood, hive bees, and foraging bees. Meanwhile, a portion is extracted or harvested as commercial products.

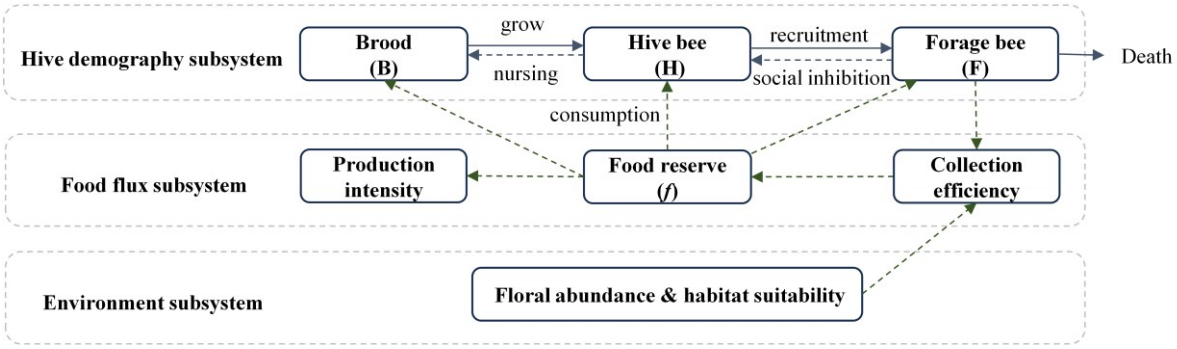


Fig. S1-1 A schematic diagram for the interaction between production operations and colony dynamics.

On this basis, colony dynamics are governed by three subsystems: (i) hive demography subsystem, characterized by age-structure and social division; (ii) food flux subsystem, governed by bees' foraging behaviors and beekeepers' production operations; (iii) stochastic environmental subsystem, defined by floral abundance and habitat suitability. In this study, an ordinary differential equation that explicitly accounts for production activities is formulated following Khoury et al. (2011; 2013). The model is based on four core states, i.e., the brood population (B), the hive bee population (H), the foraging bee population (F), and food reserves (f), and their interwind relationships. Equations governing each of these four states are provided as follows.

First, the rate of change in the brood population is given by

$$\frac{dB}{dt} = \underbrace{L(t)\frac{f^2}{b^2+f^2}\frac{H}{H+v}}_{\text{new brood}} - \underset{\text{maturation from brood to hive bees}}{\phi B} \tag{S1-1}$$

where the first term represents the number of new broods, which is related to the seasonal egg-laying rate $L(t)$, the number of hive bees in an approximated linear function $H/(H+v)$, and food reserves in a sigmoid form $f^2/(b^2+f^2)$; the second item indicates the number of broods that have transformed into adult bees.

Next, the rate of change in the hive bee population is given by

$$\frac{dH}{dt} = \underset{\text{new hive bees}}{\phi B} - \underbrace{H\left(\alpha_{\min} + \alpha_{\max}\left(\frac{b^2}{b^2+f^2}\right) - \sigma\left(\frac{F}{F+H}\right)\right)}_{\text{transition from hive bees into foraging bees by recruitment behavior}} \tag{S1-2}$$

which denotes the difference between the emergence rate of new hive bees from brood and the recruitment rate. Note that the recruitment rate $\alpha_{\min} + \alpha_{\max} b^2/(b^2+f^2) - \sigma F/(F+H)$ increases under food scarcity but decreases with the number of active foragers.

Then, the rate of change in the foraging bee population is described by Eq. (S1-3), where the first term represents newly recruited foraging bees and the second term represents the seasonal mortality rate.

$$\frac{dF}{dt} = \underbrace{H\left(\alpha_{\min} + \alpha_{\max}\left(\frac{b^2}{b^2+f^2}\right) - \sigma\left(\frac{F}{F+H}\right)\right)}_{\text{new foraging bees through recruitment behavior}} - \underbrace{m(t)F}_{death} \tag{S1-3}$$

Finally, the rate of change in food reserves is expressed as

$$\frac{df}{dt} = \underset{\text{food collection}}{cF} - \underbrace{\left(\gamma_B B + \gamma_H H + \gamma_F F\right)}_{\text{comsuption of brood, hive bees, and forgaing bees}} - \underset{\text{extracted quantity}}{\delta\beta} \tag{S1-4}$$

where c denotes the average collection efficiency per foraging bee and fluctuates according to environmental conditions; the second term corresponds to food consumption of bees across all age classes; notably, the third term accounts for the quantity of food extracted per colony by beekeepers, with parameters β and δ denoting the maximum amount of food (e.g., honey or pollen) that can be extracted and the extracted proportion relative to that maximum, respectively. Thus, beekeepers can regulate colony dynamics by adjusting the amount of bee products harvested, such as by varying the frequency of honey collection or the duration of pollen trap installation. Doing so would affect the amount of remaining food reserves in the hive, and indirectly colony mortality and brood survival, thereby shaping colony dynamics.

S1.3 Simulation experiments

This subsection depicts colony dynamics by modeling how colonies respond to different production intensities (i.e., the amount of bee products harvested) using the aforementioned mathematical model and extensive simulation experiments.

Related parameters are set based on the lifecycle of worker bees and relevant literature (Khoury et al., 2013; Russell et al., 2013). First, constant parameters include $b = 500$, $v = 5000$, $\phi = 1/21$, $\alpha_{\min} = 0.25$, $\alpha_{\max} = 0.25$, $\sigma = 0.75$, $\gamma_B = 0.018g$, $\gamma_H = \gamma_F = 0.007g$, and $\beta = 2000g$. Second, the egg-laying rate per day varies seasonally, being $L = 1500$ in spring, summer, and autumn, and $L = 0$ in winter. Additionally, the monthly natural mortality rate is scaled relative to the highest value $m = 0.102$ with a seasonal index varying chronically from January to December (i.e., 0.29, 0.29, 0.54, 0.79, 0.68, 0.57, 0.78, 1, 0.83, 0.67, 0.29, and 0.29). Third, the average collection efficiency per foraging bee is set to 0.1g per day, and is then randomly sampled from a uniform distribution in the range $[0.09, 0.11]$ in each simulation independently.

On this basis, we conducted extensive numerical experiments focusing on two core factors: initial colony strength and imposed production intensity. Specifically, the typical colony size, ranging from approximately 10000 to 55000 bees, is divided into 15 intervals $h \in \mathcal{H}$ with a step size of 3000, each with its corresponding lower and upper bounds. Similarly, the production intensity between $[0,1]$ is discretized equally into 20 grades $\ell \in \mathcal{L}_{fine}$. That is, the maximum amount of bee products harvested, estimated at 2000 grams per day per colony, is discretized into 20 grades, each representing an increment of 100 grams. For each interval of colony strength, 20 production intensities are considered across spring, summer, autumn, and winter, yielding $15 \times 20 \times 4 = 1200$ combinations. For each combination, 10000 independent simulations were performed based on the abovementioned ordinary differential equations (Eqs. (S1-1) to (S1-4)), with input of randomly generated colony strength and production intensity from their respective interval bounds, as well as foraging efficiency c within $[0.09, 0.11]$. The simulation runs for 30 days, with early termination triggered if any of the core states *B*, *H*, *F*, and *f* reaches zero. According to the 'Hou' division (i.e., 5 days), the results are recorded every five days, yielding 6 observations per simulation. Consequently, a total of 72 million ($15 \times 20 \times 4 \times 10000 \times 6 = 72000000$) pieces of data have been generated during the experiment, comprising of (i) season parameter $s \in \mathcal{S}$; (ii) the initial colony strength $(H+F)_{initial}$ and its interval index $h_{initial}$, (iii) production intensity δ and its located interval index ℓ, (iv) the

resulting colony strength $(H+F)_{resulting}$ and its associated interval index $h_{resulting}$, and (v) the difference between interval indexes of initial and resulting colony strengths, i.e., $h_{\Delta} = h_{resulting} - h_{initial}$.

To prevent the rapid deterioration of bee colonies, raw data were systematically preprocessed. For each combination with 60,000 data points, we classify and count the frequency accordingly, denoted as $N(h_{initial}, h_{\Delta}, g, \ell, s)$. Those productions, which lead to rapid deterioration (i.e., $h_{\Delta} \le -2$) or even colony collapse (i.e., early termination) to a large extent, are considered as infeasible. In other words, only those satisfying $\sum_{h_{\Delta}>-2} N(h_{initial}, h_{\Delta}, g, \ell, s) \Big/ 60000 \ge 30\%$ are deemed feasible and denoted as $\Theta(h_{initial}, g, s)$. Therefore, the average increment of colony strength $\bar{h}_{h_{initial},\ell,g,s}$ is given by

$$\bar{h}_{h_{initial},\ell,g,s} = \frac{\sum_{h_{\Delta}\ge-2} h_{\Delta} \times N(h_{initial}, h_{\Delta}, g, \ell, s)}{\sum_{h_{\Delta}\ge-2} N(h_{initial}, h_{\Delta}, g, \ell, s)}, \forall h_{initial} \in \mathcal{H}, \forall \ell \in \Theta(h_{initial}, g, s), \forall g \in \mathcal{G}, \forall s \in \mathcal{S} \quad \text{(S1-5)}$$

Fig. S1-2 and Fig. S1-3 visualize the change in the average increment in colony strength (z-axis) with varying initial colony strength (x-axis) and production intensity (y-axis) under stochastic foraging efficiencies, in the context of nectar and pollen production, respectively. Each subfigure corresponds to a specific season, and the color scale, ranging from blue to red, maps the average increase in colony strength from low to high.

According to the figures, the following observations elucidate the interplay between production operations and colony dynamics:

Observation 1: Provided that the colony can maintain homeostasis, there exists a maximum sustainable production intensity for any given colony strength. Specifically, stronger colonies exhibit a correspondingly higher threshold.

Observation 2: For any given colony strength, a negative correlation exists between the imposed production intensity and the resulting increment in colony strength. In other words, the increment of colony strength decreases with increasing production intensity, eventually turning negative (i.e., colony deterioration), a pattern consistently observed across all colony strengths. Note that the actual increment fluctuates around the reported mean values under each specific scenario.

Observation 3: For any given production intensity, the increment in colony strength demonstrates an initial increase, followed by a decreasing pattern as colony strength

increases, suggesting that more robust colonies are not necessarily more favorable for colony development.

Observation 4: Distinct seasonal patterns are observed. The colony shows the most robust enhancement in spring and summer, followed by a moderate improvement in autumn, while a typical decline occurs in winter.

Observation 5: The increased colony strength during pollen production is consistently lower than that during nectar production across all circumstances, resulting in additional physical damage and a higher mortality rate for individual bees.

These observations provide an intuitive understanding of the impacts of product category and production intensity on colony dynamics.

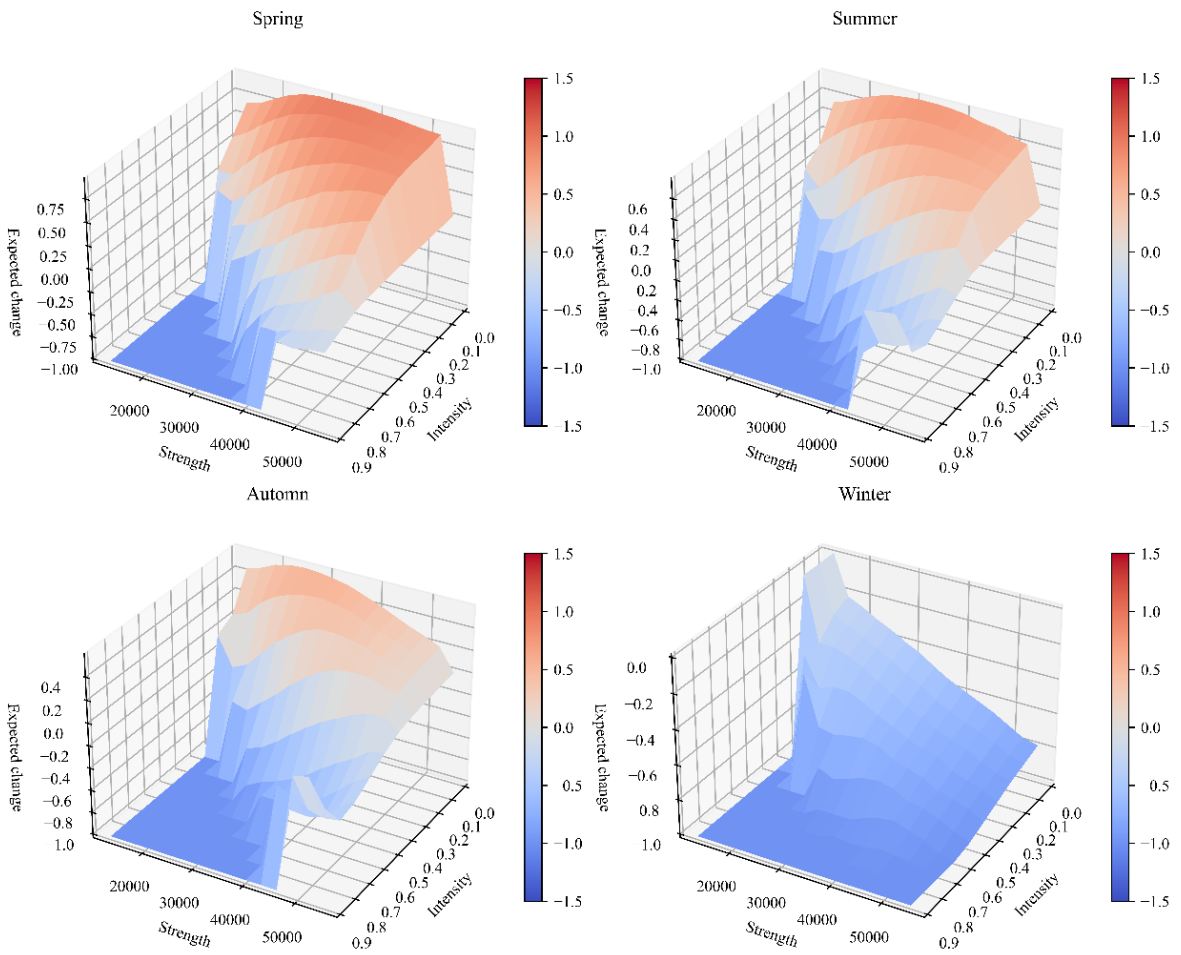


Fig. S1-2 The expected increment in colony strength in the context of nectar production

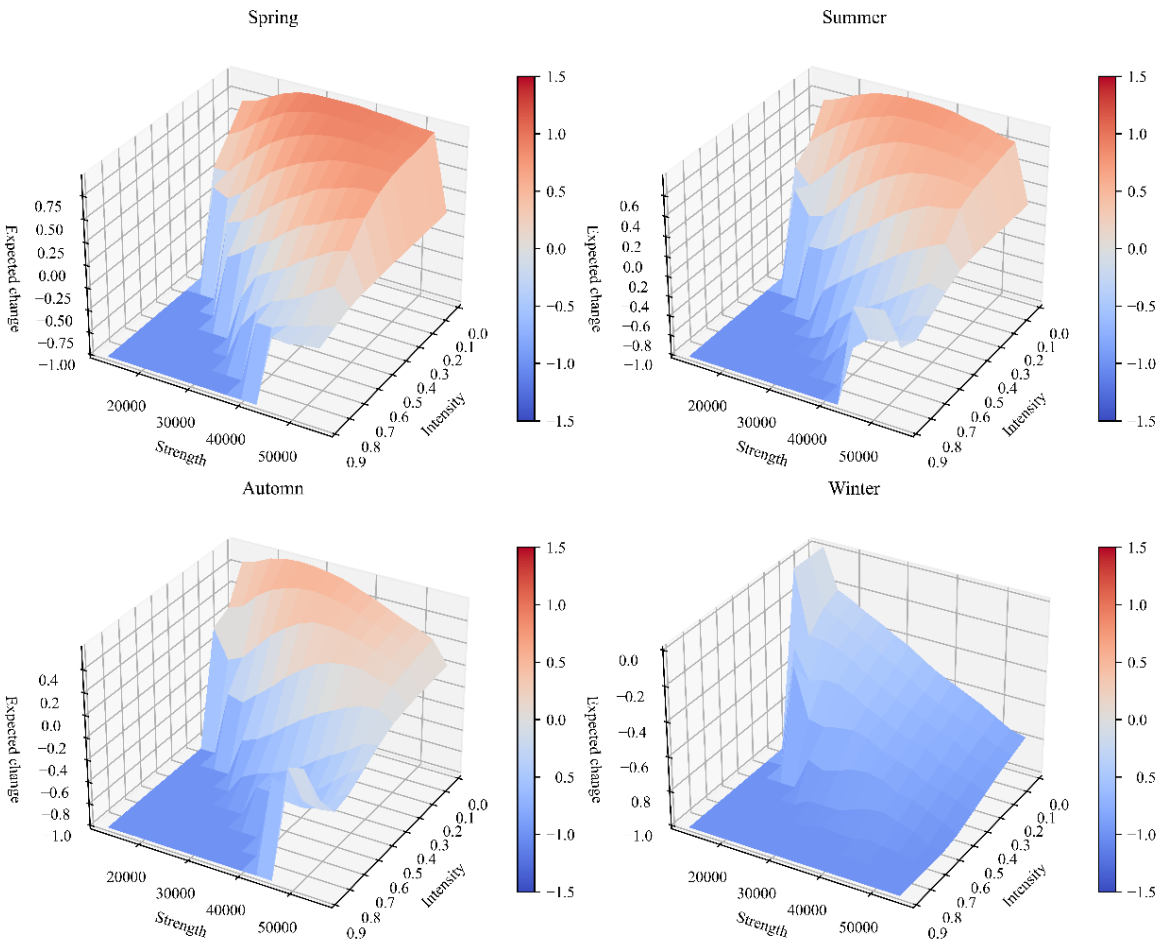


Fig. S1-3 The expected increment in colony strength in the context of pollen production

S1.4 Measurement of the transition probability of colony strength

Given the data obtained, it is unlikely to divide production intensity into such a fine discretization for practical reasons. Therefore, production intensities are handled at a coarser granularity with several predefined levels $\mathcal{L}=\{l_1,l_2,...,l_{\max}\}$ by aggregating the 20 intervals. Specifically, the aggregated level of production intensity l corresponds to $\mathcal{L}^l_{fine}=\left[\lceil 20/l_{\max}\rceil l,\max\left(\lceil 20/l_{\max}\rceil(l+1),20\right)\right]$. Let $P_{hh'lgt}$ be the probability that a colony transits from colony strength h to h' after production intensity l has been imposed on the colony in producing product category g, which can be expressed as

$$P_{hh'lgs}=\frac{\sum_{\ell\in\mathcal{L}^l_{fine}\cap\Theta(h,g,s)}\ \sum_{h_\Delta\in\left[h^L_\Delta,h^U_\Delta\right]|h+h_\Delta=h'} N\left(h,h_\Delta,g,\ell,s\right)}{\sum_{\ell\in\mathcal{L}^l_{fine}\cap\Theta(h,g,s)}\ \sum_{h_\Delta\in\left[h^L_\Delta,h^U_\Delta\right]} N\left(h,h_\Delta,g,\ell,s\right)},\forall h,h'\in\mathcal{H},\forall l\in\mathcal{L},\forall g\in\mathcal{G},\forall s\in\mathcal{S} \quad \text{(S1-6)}$$

where the denominator indicates the total occurrence of colonies with colony strength h

adopting production intensity l, while the numerator indicates the total frequency of colony transiting from level h to h' within the denominator item.

Note that if production intensity l is infeasible for colonies at level h, i.e., $\mathcal{L}^{l}_{fine} \cap \Theta(h,g,s) = \varnothing$, the value of $P_{hh'lgs}$ is 0. The above steps are repeated until the transition probability is fully derived. This study is conducted using the transition probability matrix with $l_{\max} = 3$, while the matrices based on finer ones are obtained using the same method.

Supplement S2: Notations

Indices and sets:

$t \in \mathcal{T}$: decision periods

$i, j \in \mathcal{N}$: vertices, including the origin o with its terminus e, and floral sources $\mathcal{I}$

$g \in \mathcal{G}$: bee product categories $\mathcal{G} = \{\mathcal{G}_i \mid i \in \mathcal{I}\}$, where $\mathcal{G}_i$ represents the set of bee product types that can be produced at source i

$h \in \mathcal{H}$: the discretized levels of colony strength ranging from a minimum survival threshold for hive viability h_{LB} and a maximum attainable upper bound h_{UB}, with an interval of δ, denoted as $\mathcal{H} = \{h_{LB}, h_{LB} + \delta, h_{LB} + 2\delta, ..., h_{UB}\}$

$l \in \mathcal{L}$: the discretized levels of harvest amount, namely harvest magnitude, each corresponding to a coefficient η_l relative to the production efficiency

$\tau \in \Gamma_i$: the number of periods remaining until arriving at the target floral source i, where $\Gamma_j = \left\{0, ..., \max\left(\kappa_{ij} \mid i \in \{o\} \cup \mathcal{I}\right)\right\}$

$\mathbf{P}_{gt} = \left(P_{hh'lgt}\right)_{\forall h,h' \in \mathcal{H}, \forall l \in \mathcal{L}, \forall g \in \mathcal{G}}$: the transition probability matrix of colony strength, with its component $P_{hh'lgt}$ representing the probability that a colony transits from colony strength h to h' after the production plan with product category g and harvest magnitude l at period t

$\mathbf{B}$: indicator matrix, where element $\beta_{ghlt} = 1$ represents the harvest magnitude $l \in \mathcal{L}$ for product category $g \in \mathcal{G}$ is feasible under colony strength $h \in \mathcal{H}$ in period $t \in \mathcal{T}$; 0 otherwise

Ω : the information sequence $\Omega(\omega^n) = \left\{\left(\mathbf{\Phi}^n_1, W^n_1\right), \left(\mathbf{\Phi}^n_2, W^n_2\right), ..., \left(\mathbf{\Phi}^n_{|\mathcal{T}|}, W^n_{|\mathcal{T}|}\right)\right\}$ of the ω^n sample path, which includes the exogenous conditions of floral sources

$\mathbf{\Phi}_t^n = \left(\phi_{it}^n\right)_{\forall i \in \left\{\mathcal{I} | T_i^S \le t < T_i^E\right\}}$ to measure floral abundance and habitat suitability, and the actual increment in colony strength $W_t^n = \left(w_{ihglt}^n\right)_{\forall i \in \left\{\mathcal{I} | T_i^s \le t < T_i^E\right\}, \forall h \in \mathcal{H}, \forall g \in \mathcal{G}, \forall l \in \mathcal{L}, \forall t \in \mathcal{T}}$ determined by external conditions $\mathbf{\Phi}_t^n$ and the transition probability matrix $\mathbf{P}_{gt}$

Parameters:

ψ : the number of days that each period spans

α : the colonies of honeybees

κ_{ij} : the migration duration from source i to j, and $\kappa_{ij} = d_{ij}/v$, where d_{ij} means the migration distance on arc (i, j) and v indicates the travel speed

c_{ij}^T : the migration cost from source i to j, and $c_{ij}^T = \left(c^f + c^u d_{ij}\right)\lceil \alpha/b \rceil$, where $c^f + c^u d_{ij}$ indicates the fixed cost and the distance-related travel cost for each migration, respectively, and $\lceil \alpha/b \rceil$ denotes the required number of vehicles, calculated by the number of bee colonies α and vehicle capacity b

c^H : the penalty cost for losing one frame of bees if the colony strength after overwintering is lower than that at the beginning of the current year

q_{ig} : the theoretically upper bound of production efficiency of bee product g that a standard bee colony can harvest at source i per period with the highest colony strength level

p_g : the market price of bee product g

$\left[T_i^S, T_i^E\right]$: the flowering periods of floral source i

λ : the typical survival rate of colonies after overwintering

φ_{ijht} : the change in colony strength when the colony with strength h migrates from floral source i to j during the migration and waiting (if the arrival time at source j is prior to the start time of its flowering period)

υ : the horizon visions

Decision variables:

x_{ijt} : binary variable, equals 1 if the beekeeper migrates from floral source i to j in period t, and 0 otherwise

z_{iglt} : binary variable, equals 1 if the beekeeper harvests product g with production intensity l at floral source i in period t, and 0 otherwise

State variables:

$y_{i\tau t}$: binary variable, initial state, equals 1 if the beekeeper arrives at floral source i after τ periods from period t, and 0 otherwise

ζ_{ht} : integer variable, initial state, the colony strength in period t

$\tilde{y}_{i\tau t}$: binary variable, post-decision state, equals 1 if the beekeeper arrives at N&P source i after τ periods from period t, and 0 otherwise

$\tilde{\zeta}_{ht}$: integer variable, post-decision state, the colony strength in period t

Supplement S3: The formulations in the MC and TD scenarios

S3.1 The deterministic model with full information for offline learning

For a scenario ω^n with complete information $\Omega\left(\omega^n\right)=\left\{\left(\mathbf{\Phi}_1^n,W_1^n\right),\left(\mathbf{\Phi}_2^n,W_2^n\right),...,\left(\mathbf{\Phi}_{|\mathcal{T}|}^n,W_{|\mathcal{T}|}^n\right)\right\}$, we can formulate the following posterior mixed-integer linear programming to maximize overall revenue from period t_0 to $|\mathcal{T}|$.

$$V_{t_0}\left(s_{t_0}^n\right)=\max_{a_t^n\in A_t\left(s_t^n\right)}\left(\begin{array}{l}\sum_{t\in\left[t_0,|\mathcal{T}|-1\right]}\left(\alpha\psi\sum_{i\in\mathcal{N}}\sum_{g\in\mathcal{G}_i}\sum_{l\in\mathcal{L}}p_g q_{ig}\eta_l z_{iglt}^n-\sum_{i\in\mathcal{I}\cup\{o\}}\sum_{j\in\mathcal{I}\cup\{e\}}c_{ij}x_{ijt}^n\right)\\+\max\left(\alpha c^H\left(\sum_{h\in\mathcal{H}}h\zeta_{ht_0}^n-\left\lfloor\lambda\sum_{h\in\mathcal{H}}h\zeta_{h|\mathcal{T}|}^n\right\rfloor\right),0\right)\end{array}\right) \tag{S3-1}$$

s.t.

$$\sum_{g\in\mathcal{G}}\sum_{l\in\mathcal{L}}z_{iglt}^n+\sum_{j\in\mathcal{I}\cup\{e\}|t+\kappa_{ij}\le T_j^E}x_{ijt}^n\le y_{i0t}^n,\quad\forall i\in\mathcal{I}\cup\{o\},\forall t\in\left[t_0,|\mathcal{T}|\right] \tag{S3-2}$$

$$\sum_{g\in\mathcal{G}}\sum_{l\in\mathcal{L}}z_{iglt}^n\le y_{i0t}^n f\left(T_i^S,T_i^E,t\right),\quad\forall i\in\mathcal{I},\forall t\in\left[t_0,|\mathcal{T}|\right] \tag{S3-3}$$

$$\sum_{i\in\mathcal{I}}z_{iglt}^n\le\sum_{h\in\mathcal{H}}\beta_{ghlt}\zeta_{ht}^n,\quad\forall g\in\mathcal{G},\forall l\in\mathcal{L},\forall t\in\left[t_0,|\mathcal{T}|\right] \tag{S3-4}$$

$$\sum_{h\in\mathcal{H}}\zeta_{ht}^n\le 1,\quad\forall t\in\left[t_0,|\mathcal{T}|\right] \tag{S3-5}$$

$$\tilde{y}_{i0t}^n\le y_{i0t}^n-\sum_{j\in\mathcal{I}\cup\{e\}|t+\kappa_{ij}\le T_j^E}x_{ijt}^n,\quad\forall i\in\mathcal{I}\cup\{o\},\forall t\in\left[t_0,|\mathcal{T}|\right] \tag{S3-6}$$

$$\tilde{y}_{j\tau t}^n\le y_{j\tau t}^n+\sum_{i\in\mathcal{I}\cup\{o\}|\kappa_{ij}=\tau}x_{ijt}^n,\quad\forall j\in\mathcal{I}\cup\{e\},\forall\tau\in\Gamma_j\mid\tau\le T_j^E-t,\forall t\in\left[t_0,|\mathcal{T}|\right] \tag{S3-7}$$

$$y_{i\tau,t+1}^n\le\begin{cases}\tilde{y}_{i\tau t}^n, & \forall\tau=0\\ \tilde{y}_{i,\tau+1,t}^n, & \forall\tau\in\Gamma_i\end{cases},\quad\forall i\in\mathcal{N},\quad\forall t\in\left[t_0,|\mathcal{T}|-1\right] \tag{S3-8}$$

$$\sum_{h\in\mathcal{H}} h\zeta_{h,t+1}^{n} \le h + \sum_{i\in\mathcal{I}}\sum_{g\in\mathcal{G}}\sum_{l\in\mathcal{L}} w_{ihglt}^{n}\zeta_{ht}^{n} z_{iglt}^{n} + \sum_{j\in\mathcal{I}\cup\{e\}|t+\kappa_{ij}\le T_j^E}\ \sum_{i\in\mathcal{I}\cup\{o\}} \sigma_{ijh} x_{ijt}^{n} + \left(1-\zeta_{ht}^{n}\right)M,$$
$$\forall h\in\mathcal{H}, \forall t\in\left[t_0, |\mathcal{T}|-1\right] \tag{S3-9}$$

$$y_{e0|\mathcal{T}|}^{n} = 1 \tag{S3-10}$$

$$x_{ijt}^{n} \in \{0,1\},\ \ \forall i\in\mathcal{I}\cup\{o\}, \forall j\in\mathcal{I}\cup\{e\}, \forall t\in\left[t_0, |\mathcal{T}|\right] \tag{S3-11}$$

$$z_{iglt}^{n} \in \{0,1\},\ \ \forall i\in\mathcal{I}, \forall g\in\mathcal{G}, \forall l\in\mathcal{L}, \forall t\in\left[t_0, |\mathcal{T}|\right] \tag{S3-12}$$

$$y_{i\tau t}^{n}, \tilde{y}_{i\tau t}^{n} \in \{0,1\},\ \ \forall i\in\mathcal{N}, \forall \tau\in\Gamma_i, \forall t\in\left[t_0, |\mathcal{T}|\right] \tag{S3-13}$$

$$\zeta_{ht}^{n} \in \{0,1\},\ \ \forall h\in\mathcal{H}, \forall t\in\left[t_0, |\mathcal{T}|\right] \tag{S3-14}$$

S3.2 The dynamic model with partial information for online learning

For a scenario ω^n, only the information $\left(\mathbf{\Phi}_{t_c}^{n}, W_{t_c}^{n}\right), \left(\mathbf{\Phi}_{t_c+1}^{n}, W_{t_c+1}^{n}\right)..., \left(\mathbf{\Phi}_{t_c+\upsilon}^{n}, W_{t_c+\upsilon}^{n}\right)$ within the visions from the current period t_c to $t_c+\upsilon$ is observed, and the online model optimizes the short-term revealed information and the long-term anticipation as follows:

$$\begin{cases} \max\limits_{a_t^n\in A_t\left(s_t^n\right)} \left(\sum\limits_{t\in\left[t_c, t_c+\upsilon\right]} R_t\left(s_t^n, a_t^n\right) + \bar{V}_{t_c+\upsilon}\left(s_{t_c+\upsilon}^{n,a}\right) \right), & t_c < |\mathcal{T}|-\upsilon \\ \max\limits_{a_t^n\in A_t\left(s_t^n\right)} \left(\sum\limits_{t\in\left[t_c, |\mathcal{T}|\right]} R_t\left(s_t^n, a_t^n\right) \right), & t_c \ge |\mathcal{T}|-\upsilon \end{cases} \tag{S3-15}$$

s.t.

$$\sum_{g\in\mathcal{G}}\sum_{l\in\mathcal{L}} z_{iglt}^{n} + \sum_{j\in\mathcal{I}\cup\{e\}|t+\kappa_{ij}\le T_j^E} x_{ijt}^{n} \le y_{i0t}^{n},\ \ \forall i\in\mathcal{I}\cup\{o\}, \forall t\in\left[t_c, \min\left(t_c+\upsilon, |\mathcal{T}|\right)\right] \tag{S3-16}$$

$$\sum_{g\in\mathcal{G}}\sum_{l\in\mathcal{L}} z_{iglt}^{n} \le y_{i0t}^{n} f\left(T_i^S, T_i^E, t\right),\ \ \forall i\in\mathcal{I}, \forall t\in\left[t_c, \min\left(t_c+\upsilon, |\mathcal{T}|\right)\right] \tag{S3-17}$$

$$\sum_{i\in\mathcal{I}} z_{iglt}^{n} \le \sum_{h\in\mathcal{H}} \beta_{ghlt}\zeta_{ht}^{n},\ \ \forall g\in\mathcal{G}, \forall l\in\mathcal{L}, \forall t\in\left[t_c, \min\left(t_c+\upsilon, |\mathcal{T}|\right)\right] \tag{S3-18}$$

$$\sum_{h\in\mathcal{H}} \zeta_{ht}^{n} \le 1,\ \ \forall t\in\left[t_c, \min\left(t_c+\upsilon, |\mathcal{T}|\right)\right] \tag{S3-19}$$

$$\tilde{y}_{i0t}^{n} \le y_{i0t}^{n} - \sum_{j\in\mathcal{I}\cup\{e\}|t+\kappa_{ij}\le T_j^E} x_{ijt}^{n},\ \ \forall i\in\mathcal{I}\cup\{o\}, \forall t\in\left[t_c, \min\left(t_c+\upsilon, |\mathcal{T}|\right)\right] \tag{S3-20}$$

$$\tilde{y}_{j\tau t}^{n} \le y_{j\tau t}^{n} + \sum_{i\in\mathcal{I}\cup\{o\}|\kappa_{ij}=\tau} x_{ijt}^{n},\ \ \forall j\in\mathcal{I}\cup\{e\}, \forall \tau\in\Gamma_j \mid \tau\le T_j^E - t, \forall t\in\left[t_c, \min\left(t_c+\upsilon, |\mathcal{T}|\right)\right] \tag{S3-21}$$

$$\sum_{h'\in\mathcal{H}} h'\tilde{\zeta}^{n}_{h',t_c+\upsilon} \le h+\sum_{g\in\mathcal{G}}\sum_{l\in\mathcal{L}}\sum_{i\in\{i|g\in\mathcal{G}_i\}}\left[\sum_{h''\in\mathcal{H}}(h''-h)\mathrm{P}_{hh''lg}\right]z^{n}_{i,g,l,t_c+\upsilon}+\sum_{j\in\mathcal{I}\cup\{e\}|t_c+\upsilon+\kappa_{ij}\le T_j^E}\sum_{i\in\mathcal{I}\cup\{o\}}\sigma_{ijh}x^{n}_{i,j,t_c+\upsilon}$$
$$+\left(1-\zeta^{n}_{h,t_c+\upsilon}\right)M,\forall h\in\mathcal{H} \tag{S3-22}$$

$$\sum_{h\in\mathcal{H}}\tilde{\zeta}^{n}_{ht}\le 1,\ \ \forall t\in\left[t_c,\min\left(t_c+\upsilon,|\mathcal{T}|\right)\right] \tag{S3-23}$$

$$y^{n}_{i,\tau,t+1}\le\begin{cases}\tilde{y}^{n}_{i\tau t}, & \forall\tau=0\\ \tilde{y}^{n}_{i,\tau+1,t}, & \forall\tau\in\Gamma_i\end{cases},\ \ \forall i\in\mathcal{N},\ \ \forall t\in\left[t_c,\min\left(t_c+\upsilon,|\mathcal{T}|\right)\right] \tag{S3-24}$$

$$\sum_{h\in\mathcal{H}}h\zeta^{n}_{h,t+1}\le h+\sum_{i\in\mathcal{I}}\sum_{g\in\mathcal{G}}\sum_{l\in\mathcal{L}}w^{n}_{ihglt}\zeta^{n}_{ht}z^{n}_{iglt}+\sum_{j\in\mathcal{I}\cup\{e\}|t+\kappa_{ij}\le T_j^E}\sum_{i\in\mathcal{I}\cup\{o\}}\sigma_{ijh}x^{n}_{ijt}+\left(1-\zeta^{n}_{ht}\right)M,$$
$$\forall h\in\mathcal{H},\forall t\in\left[t_c,\min\left(t_c+\upsilon,|\mathcal{T}|\right)\right] \tag{S3-25}$$

$$y^{n}_{e0|\mathcal{T}|}=1 \tag{S3-26}$$

$$x^{n}_{ijt}\in\{0,1\},\ \ \forall i\in\mathcal{I}\cup\{o\},\forall j\in\mathcal{I}\cup\{e\},\forall t\in\left[t_c,\min\left(t_c+\upsilon,|\mathcal{T}|\right)\right] \tag{S3-27}$$

$$z^{n}_{iglt}\in\{0,1\},\ \ \forall i\in\mathcal{I},\forall g\in\mathcal{G},\forall l\in\mathcal{L},\forall t\in\left[t_c,\min\left(t_c+\upsilon,|\mathcal{T}|\right)\right] \tag{S3-28}$$

$$y^{n}_{i\tau t},\tilde{y}^{n}_{i\tau t}\in\{0,1\},\ \ \forall i\in\mathcal{N},\forall\tau\in\Gamma_i,\forall t\in\left[t_c,\min\left(t_c+\upsilon,|\mathcal{T}|\right)\right] \tag{S3-29}$$

$$\zeta^{n}_{ht},\tilde{\zeta}^{n}_{ht}\in\{0,1\},\ \ \forall h\in\mathcal{H},\forall t\in\left[t_c,\min\left(t_c+\upsilon,|\mathcal{T}|\right)\right] \tag{S3-30}$$

The majority of constraints remain the same with the posterior model, but within the visions rather than the annual cycle. When $t_c<|\mathcal{T}|-\upsilon$, the objective function successively accumulates the revenue $R_t\left(s^n_t,a^n_t\right)$ within the horizon visions, and adds post-decision approximations, where $R_t\left(s^n_t,a^n_t\right)$ defines the difference between income and migration costs. If $t_c\ge|\mathcal{T}|-\upsilon$, the planning period changes from $\left[t_c,t_c+\upsilon\right]$ to $\left[t_c,|\mathcal{T}|\right]$, then the piecewise objective function includes the accumulated revenue and possible penalty.

Supplement S4: Supplement to experimental results

S4.1 Details of instance generation

The instances are based on discretized colony strength and production intensity according to the "Hou" division (5 days), which helps guide agricultural activities. Specifically, colony strength, ranging from 10000 to 55000, is categorized into 15 grades, each increment of 3000 bees, roughly the capacity of a single comb, allowing practical field estimation via comb counts. Production intensity, defined as the ratio of actual yield to the production efficiency per colony

in each period (bounded in [0,1]), is equably discretized into three levels (low, medium, and high), with benefit coefficients as the midpoints between adjacent levels, i.e., $\eta_l = (l-1+l)/(2|\mathcal{L}|)$.

In each episode, environmental conditions are uniformly distributed over [0,1], serving as a composite index of floral abundance and habitat suitability. Then, the actual increment in colony strength is derived via the transition probability matrix and environmental conditions. For example, if a bee colony with strength h produces product category g with production intensity l in period t, it may evolve into strength h_1, h_2, and h_3 (in ascending order) with probabilities P_{hh_1lgt}, P_{hh_2lgt}, and P_{hh_3lgt}, respectively. The resulting strength is determined by which interval (i.e., $\left[0, P_{hh_1lgt}\right]$, $\left(P_{hh_1lgt}, P_{hh_1lgt}+P_{hh_2lgt}\right]$, and $\left(P_{hh_1lgt}+P_{hh_2lgt}, 1\right]$) the random number falls into. This process iterates to generate all information of each realization, yielding a four-dimensional matrix indexed by colony strength, product category, production intensity, and corresponding season.

Additionally, beekeeper-related parameters are homogeneous across origins, including the number of colonies $\alpha = 150$ and the initial colony strength $h_0 = 1$ among the 15 grades. Other parameters are set as follows: the horizon vision $\upsilon = 9$, the penalty cost per frame $c^H = 20$, and the survival rate after overwintering $\lambda = 0.8$.

S4.2 Comparative results for cases originating from origins O2, O3, and O4

Table S4-1 Comparison of KPIs across different policies based on instances originating from O2.

N	Policy	Rev.	Imp. (%)	Gap (%)	Unit rev.	Unit income	Unit cost	#Source	Prod. period	Avg. str.	Avg. int.
25	Posterior-RP	37118.19	-	-	247.45	353.81	106.36	9.28	43.85	8.02	2.49
	ADP	35852.78	-	4.23	239.02	346.72	107.70	9.38	43.67	8.04	2.46
	Static-R/Rule-based-P	21029.37	42.58	-	140.20	241.81	101.61	9.00	44.20	4.42	1.91
	Static-R/Greedy-P	22637.20	37.05	-	150.91	252.53	101.61	9.00	44.20	1.75	1.95
	Rolling-RP	13212.94	64.42	-	88.09	156.93	68.84	5.91	26.18	3.77	2.08
50	Posterior-RP	45722.29	-	-	304.82	459.07	154.25	9.67	47.49	9.12	2.67
	ADP	44478.19	-	3.25	296.52	453.79	157.27	9.84	47.46	9.16	2.64
	Static-R/Rule-based-P	22401.35	51.19	-	149.34	288.78	139.44	8.70	46.90	4.71	1.92
	Static-R/Greedy-P	24511.76	45.94	-	163.41	302.85	139.44	8.70	46.90	1.91	1.98
	Rolling-RP	24061.26	49.89	-	160.41	278.97	118.56	7.68	34.08	7.26	2.37
100	Posterior-RP	66671.02	-	-	444.47	598.49	154.02	10.75	53.57	9.66	2.77
	ADP	65824.33	-	1.33	438.83	593.56	154.73	10.78	53.40	9.77	2.75

Static-R/Rule-based-P	36155.65	41.39	-	241.04	378.50	137.46	9.70	53.40	5.24	2.01
Static-R/Greedy-P	39739.58	36.82	-	264.93	402.39	137.46	9.70	53.40	2.84	2.07
Rolling-RP	43131.10	52.93	-	287.54	427.04	139.50	10.10	43.85	9.44	2.57

Table S4-2 Comparison of KPIs across different policies based on instances originating from O3.

N	Policy	Rev.	Imp. (%)	Gap (%)	Unit rev.	Unit income	Unit cost	#Source	Prod. period	Avg. str.	Avg. int.
25	Posterior-RP	38835.86	-	-	258.91	357.20	98.29	9.57	44.28	8.01	2.49
	ADP	37606.30	-	3.89	250.71	349.94	99.23	9.61	44.02	8.08	2.46
	Static-R/Rule-based-P	22173.72	41.39	-	147.82	242.36	94.54	9.20	44.80	4.48	1.90
	Static-R/Greedy-P	23576.52	36.82	-	157.18	251.72	94.54	9.20	44.80	1.60	1.94
	Rolling-RP	17486.40	52.93	-	116.58	179.32	62.75	6.48	28.57	4.23	2.10
50	Posterior-RP	48686.46	-	-	324.58	465.85	141.28	10.32	48.36	9.16	2.67
	ADP	47358.37	-	3.39	315.72	456.41	140.69	10.33	47.97	9.10	2.64
	Static-R/Rule-based-P	25118.60	49.27	-	167.46	291.13	123.67	9.10	47.60	4.66	1.92
	Static-R/Greedy-P	27780.27	42.55	-	185.20	308.88	123.67	9.10	47.60	2.00	1.98
	Rolling-RP	27872.91	45.01	-	185.82	294.78	108.96	8.28	36.25	7.43	2.38
100	Posterior-RP	67280.09	-	-	448.53	599.62	151.08	10.83	53.61	9.66	2.77
	ADP	66476.98	-	1.25	443.18	594.99	151.81	10.87	53.55	9.76	2.75
	Static-R/Rule-based-P	35760.25	46.66	-	238.40	377.22	138.82	10.00	53.90	5.22	1.98
	Static-R/Greedy-P	40648.58	39.65	-	270.99	409.81	138.82	10.00	53.90	2.95	2.08
	Rolling-RP	44967.20	34.11	-	299.78	436.73	136.95	10.18	44.70	9.57	2.58

Table S4-3 Comparison of KPIs across different policies based on instances originating from O4.

N	Policy	Rev.	Imp. (%)	Gap (%)	Unit rev.	Unit income	Unit cost	#Source	Prod. period	Avg. str.	Avg. int.
25	Posterior-RP	37048.82	-	-	246.99	353.10	106.11	9.34	43.72	8.02	2.49
	ADP	35687.12	-	4.85	237.91	347.02	109.11	9.49	43.96	8.01	2.45
	Static-R/Rule-based-P	20580.99	43.96	-	137.21	233.54	96.33	8.90	42.80	4.23	1.88
	Static-R/Greedy-P	22389.34	38.07	-	149.26	245.60	96.33	8.90	42.80	1.59	1.94
	Rolling-RP	10292.49	73.79	-	68.62	123.00	54.39	4.91	21.70	2.44	2.01
50	Posterior-RP	47648.99	-	-	317.66	463.07	145.41	10.05	47.78	9.12	2.66
	ADP	46496.28	-	2.87	309.98	455.59	145.61	10.07	47.54	9.16	2.65
	Static-R/Rule-based-P	24410.07	49.03	-	162.73	290.43	127.69	8.90	46.60	4.80	1.94
	Static-R/Greedy-P	26603.80	44.40	-	177.36	305.05	127.69	8.90	46.60	1.98	1.99
	Rolling-RP	23180.23	53.93	-	154.53	263.77	109.24	7.61	32.98	6.75	2.33
100	Posterior-RP	69378.70	-	-	462.52	602.32	139.80	11.06	53.93	9.61	2.76
	ADP	68543.08	-	1.28	456.95	596.97	140.01	11.08	53.84	9.69	2.74
	Static-R/Rule-based-P	38193.16	44.93	-	254.62	382.93	128.31	10.30	54.20	5.32	1.99
	Static-R/Greedy-P	43119.66	38.01	-	287.46	415.77	128.31	10.30	54.20	3.11	2.09
	Rolling-RP	48512.03	30.56	-	323.41	453.97	130.56	10.53	45.86	9.65	2.59

S4.3 Visualization of migration routes originating from other origins

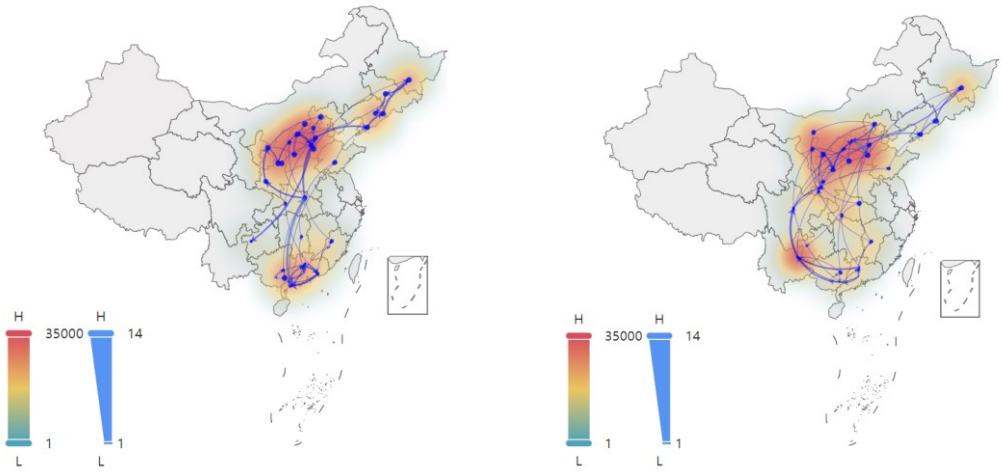


Fig. S4-1 Visualization of migration routes originating from "Guangdong" for nectar-only (Left) and adaptive production (Right)

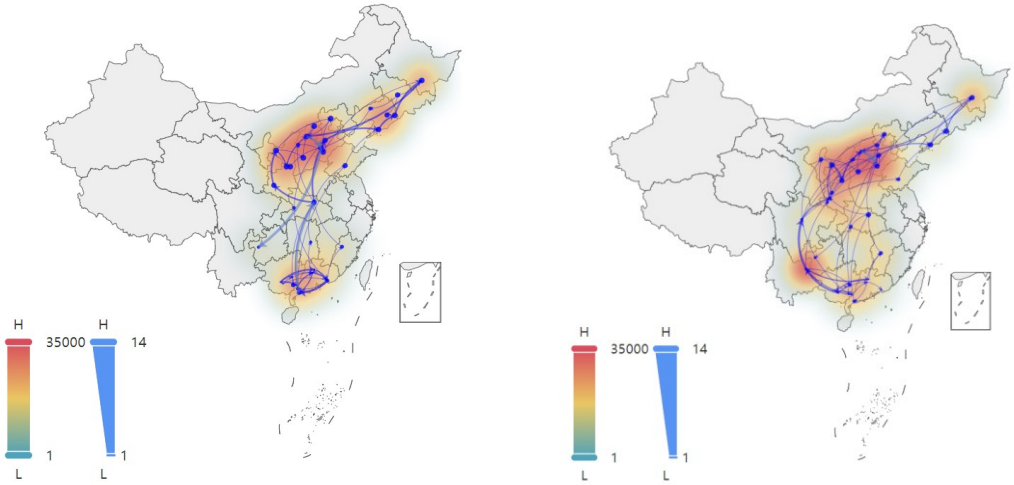


Fig. S4-2 Visualization of migration routes originating from "Guangxi" for nectar-only (Left) and adaptive production (Right)

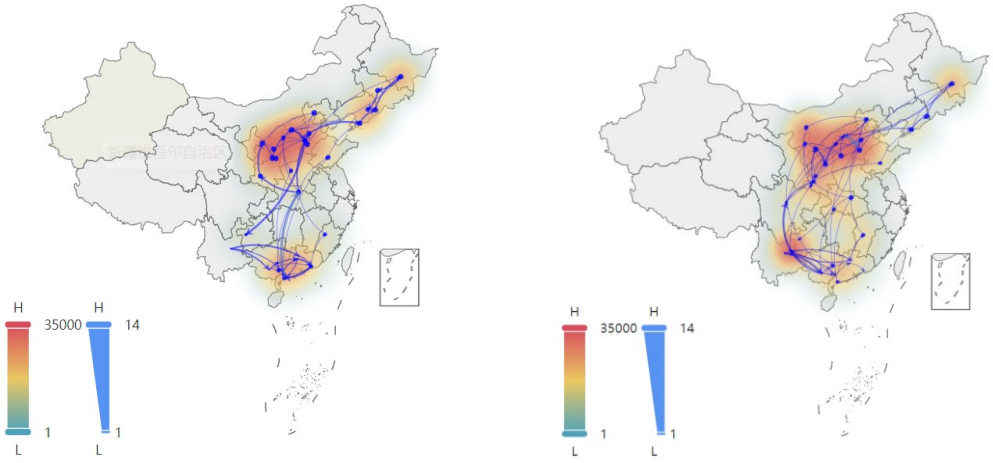


Fig. S4-3 Visualization of migration routes originating from "Yunnan" for nectar-only (Left) and adaptive production (Right)

S4.4 Supplement to sensitivity analysis

This subsection conducts sensitivity analyses on the following parameters: the number of production intensity levels $|\mathcal{L}|$, the initial strength of bee colonies h_0, and the number of bee colonies. We vary the values of these parameters individually (i.e., $|\mathcal{L}|$ changed from 3 to 7, h_0 changed from 1 to 5 within 15 grades, and α ranged from 100 to 300 in increments of 50) to assess their impact on the annual revenue (Fig. S4-4) and other KPIs (Fig. S4-5). The results show that beekeepers' personal characteristics substantially affect overall revenue and KPI performance as follows.

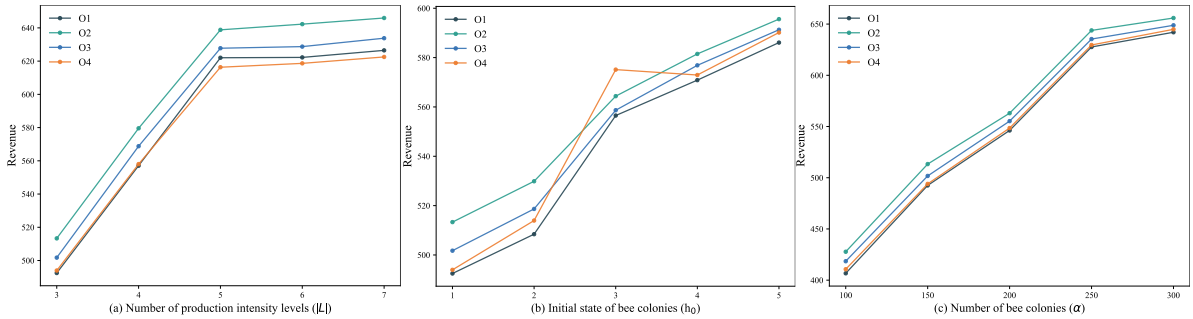
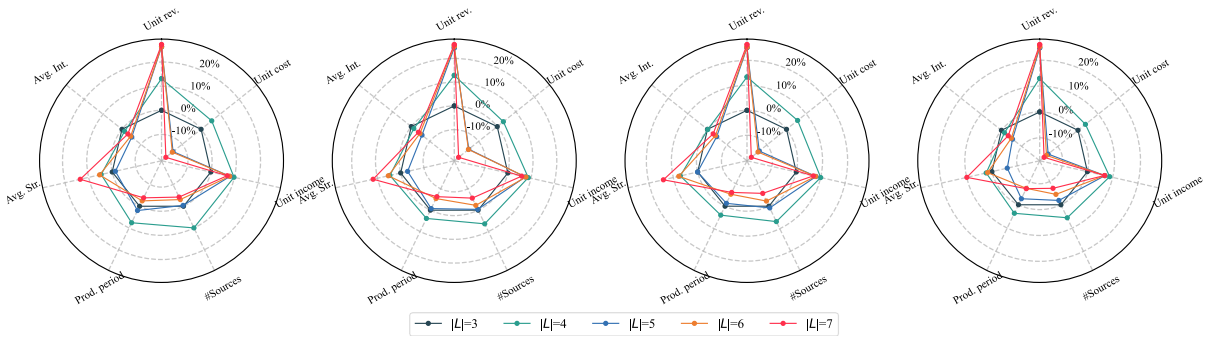
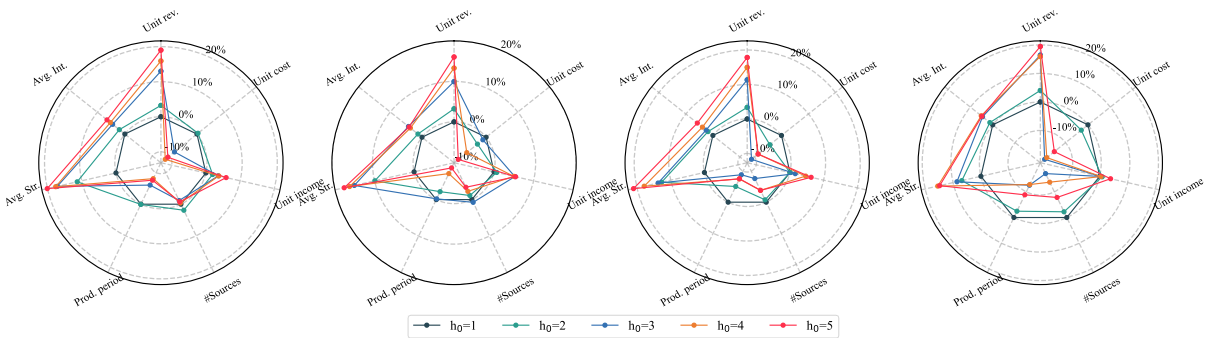


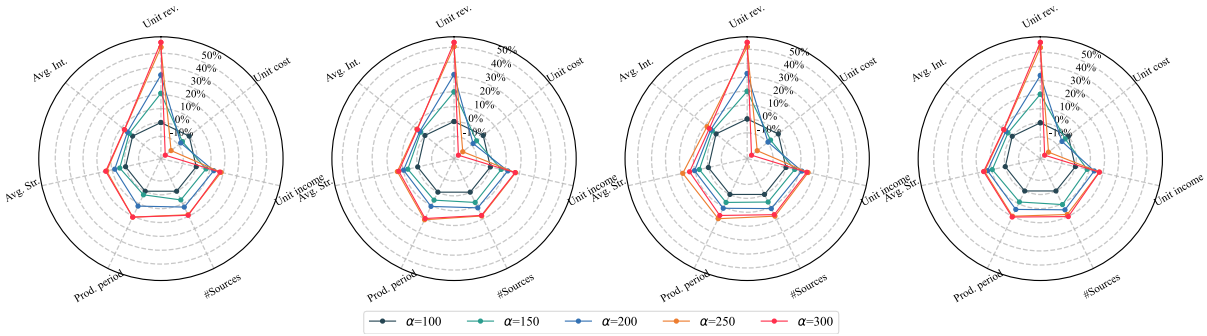
Fig. S4-4 Impact of a beekeeper's personalized characteristics on the annual revenue



(a)



(b)



(c)

Fig. S4-5 Impact of a beekeeper's personalized characteristics on other KPIs

- **Impact of the number of production intensity levels**. Fig. S4-4(a) and Fig**. S4-5**(a) show the trends of beekeeping revenue and other KPIs as $|\mathcal{L}|$ are progressively refined from coarse- to fine-grained levels. The results reveal that beekeeping revenue increases with more refined production management, but with diminishing marginal returns. At the coarse granularity, beekeeping revenue is driven by extended production periods ("Prod. period") and high intensity levels ("Avg. int."). However, it incurs substantial migration costs due to frequent migration to high-quality floral sources for production or reproduction. In contrast, finer granularity naturally facilitates precise regulation and utilization of colony strength, significantly reducing dependence on reproductive-supporting sources and improving the flexibility of migration routes, thereby reducing migration costs and improving annual revenue. However, this improvement is quite limited beyond a certain threshold.
- **Impact of the initial strength of bee colonies**. The initial strength of bee colonies has a clear positive effect on beekeeping revenue, as illustrated in Fig. S4-4(b) and Fig**. S4-5**(b). On the one hand, higher initial strength promotes colony growth, enabling intensive production and rising revenues. However, this increased income is restrained by the premature termination of production to preserve sufficient colony strength for the coming year, as evidenced by indicators "#Sources" and "Prod. period". On the other hand, increased initial strength reduces migration costs by fewer migration events and shorter migration distances. This is because the enhanced production potential of stronger colonies reduces reliance on specific floral sources for strength maintenance or improvements, enabling broader alternatives and more flexible route scheduling. In conclusion, stronger bee colonies at origins naturally enhance average colony strength and boost income through intensive production, but, more importantly, through flexible floral source selection and efficient migratory routing.

- **Impact of the number of bee colonies.** Fig. S4-4(c) and Fig**. S4-5**(c) illustrate the impacts of colony number on beekeeping revenue and other KPIs, in which the migration costs asscoaited with migration linkes are correspondingly adjusted based on beehive numbers. Fig. S4-4(c) shows a significant upward trend in beekeeping revenue with increasing numbers of beehives, consistent with the expectation of economies of scale. Notably, the beekeeping revenue per colony also improves by 20% to 50% compared to the baseline of 100 beehives, as shown in Fig. S4-5(c). As the number of colonies increases, migration costs per colony decline and production potential rises, diminishing the relative importance of migration costs in routing decisions. It improves production performance by longer migration routes and extended production periods, and prioritizes access to high-quality floral sources, even when longer travel distances and higher costs are incurred. Conversely, few colonies should strike a balance between production income and migratory costs, resulting in more geographically constrained migration routes.

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